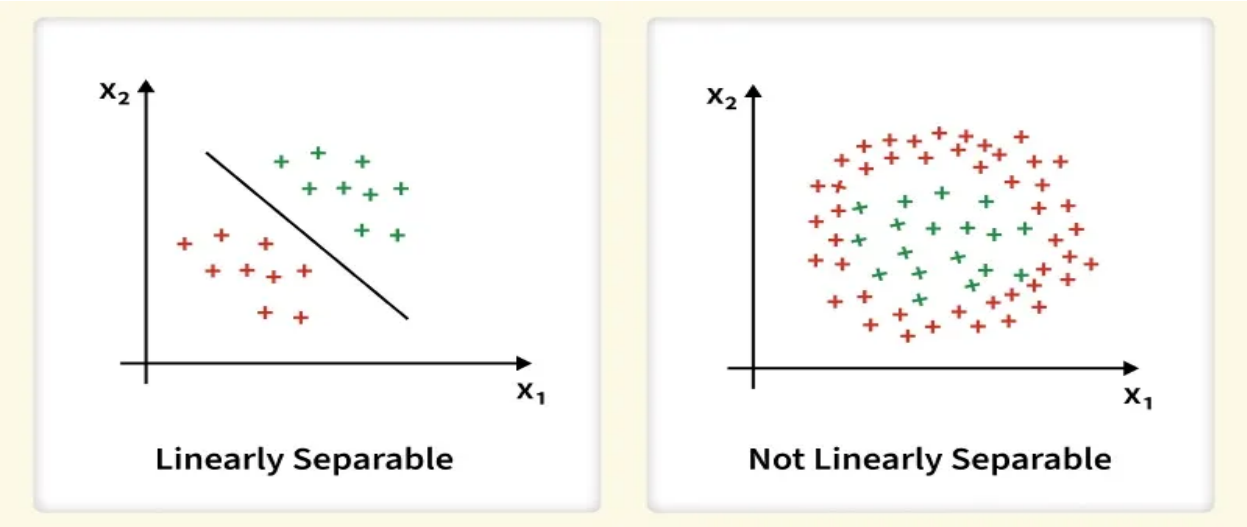
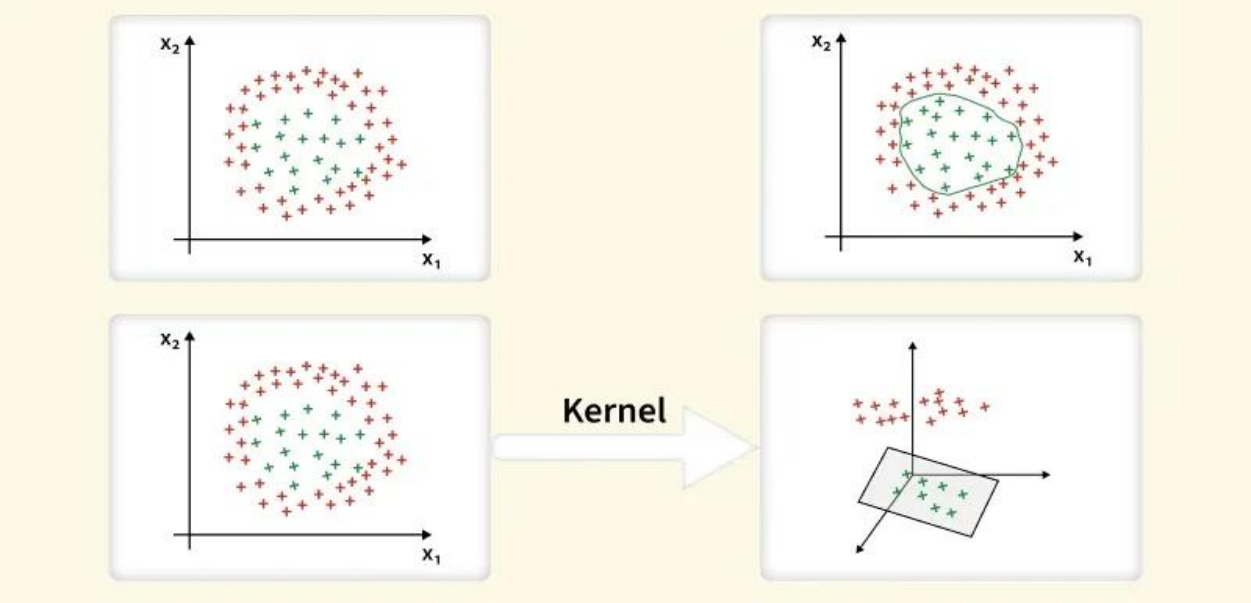
Svm kernels

Kernel Function is a method used to take data as input and transform it into the required form of processing data. It computes how similar two points look after being projected into a higher feature space, without ever performing the projection.





Some of the ideas behind kernels in SVM are:

1. Similarity Measurement: Kernels calculate how closely two data points are related in feature space.
2. Dimensional Expansion: They map data into higher dimensions implicitly without expensive computation.
3. Flexible Boundaries: They enable the model to draw curved or irregular separation lines.
4. Efficient Computation: They avoid manually engineering additional feature columns.
5. Inner Product Trick: Kernels compute complex feature interactions using simple mathematical shortcuts.

Using Kernel Functions

Some of the reasons to use kernel functions are:

1. Non-Linear Patterns: Many real-world problems cannot be separated with straight lines.
2. Smooth Separation: Kernels allow boundaries that curve around data clusters.
3. Reduced Feature Work: They eliminate manual creation of polynomial or cross terms.
4. Improved Accuracy: Non-linear transformations capture deeper relationships.
5. General Adaptability: Kernels make SVM work well across diverse data types.

Mathematical Formula

Mathematically, SVM Kernel Function can be represented by:

$$K(x, x') = \phi(x)^{\top}\phi(x')$$

Linear Kernel

Measures similarity using a simple dot product; best for linearly separable data with fast computation.

$$K(x, x') = x^\top x'$$

Characteristics

1. **Dot Product Similarity:** It evaluates similarity using a straightforward dot product between feature vectors.
2. **Straight Boundaries:** It works best when the data is linearly separable.
3. **Fast Training:** It trains faster compared to more complex kernels.
4. **Sparse Features:** It is commonly used for text or document classification tasks.
5. **Wide Margins:** It produces boundaries that remain clean and interpretable.

Polynomial Kernel

Captures more complex relationships by raising feature interactions to a power, allowing curved decision boundaries.

$$K(x, x') = (x^\top x' + r)^d$$

Characteristics

1. **Feature Interactions:** It represents combinations of original features up to a selected power.
2. **Degree Control:** Higher degrees create more twists and turns in the decision boundary.
3. **Smooth Curvature:** It handles patterns that bend gently across feature space.

4. **Overfitting Risk:** Improper degree settings may cause the model to memorize noise.
5. **Adjustable Shape:** Coefficients allow fine-grained boundary customization.

RBF (Gaussian) Kernel

Maps data into infinite dimensions and works well when class boundaries are highly non-linear and irregular.

$$K(x, x') = e^{-\gamma|x-x'|^2}$$

Characteristics

1. **Distance-Based Similarity:** Closer points are treated as more alike, forming tight clusters.
2. **Flexible Curves:** It adapts to boundaries that wrap and spiral around complex shapes.
3. **Gamma Influence:** Higher gamma values make the decision boundary more sensitive.
4. **Default Choice:** It performs well when data complexity is unknown.
5. **Smooth Output:** It constructs continuous and visually appealing separation surfaces.

- ***References:*** ***https://www.geeksforgeeks.org/machine-learning/major-kernel-functions-in-support-vector-machine-svm/***